Interface Task Selection & Empirical Problem Framing

Target System Flow: For my experiment, I will be testing to see if my redesigned Early Voting interface from a previous assignment improves usability compared to the original design on the Madison County Elections website.

Empirical Rationale: The bottleneck that I'm going to be testing is how long it takes a user to find information (operational hours, address) of an Early Voting location that they'd like to use. The Early Voting interface I've already redesigned should have multiple positives compared to the old design. It should minimize cognitive load by not bombarding the user with the entire list of every single Early Voting location, but rather revealing it to them in a drop down menu that is better sorted than the current list. It should also optimize task completion time as now that the list is better sorted than the original, an average user should be able to find their desired early voting location more quickly in the list of locations, optimizing task completion time.

Hypothesis Formulation

Directional Hypotheses:

1. Users completing the early voting information gathering task with interface B (my redesigned prototype) will have a lower average task completion time than users completing the same task with interface A (the original interface on the Madison County Elections website)
2. Users completing the task with interface B will report higher average ease-of-use ratings than when using interface A

Null Hypotheses:

1. Interface B will not produce a lower average task completion time than interface A
2. Interface B will not produce higher average ease-of-use ratings than interface A.

Variable Formulation & Operationalization

Indepndent Variables (IVs):

1. *Interface Version* - interface A v.s. interface B

Dependent Variables (DVs):

1. *Task Completion Time* - Seconds from starting the task until the correct early voting address and hours are identified
2. *Error Count* - Number of incorrect selections or navigation actions
3. *Ease-of-Use-Rating* - Subjective 1-7 rating after task is complete, 1 being the task was very difficult to complete, 7 being the task was very easy to complete, 4 being neutral.

Confounding Variables & Controls:

1. *Previous Website Experience* - We must ask whether participants have used the Madison County Elections website before
2. *Device and Browser* - We must run all of the experiments on the same computer, using the same browser, with the same keyboard/mouse, and with the same initial setup (i.e. is the browser already opened to a blank page? Are we just on the desktop? Is interface A or B already visible?)
3. *Testing Environment* - We must run the experiments in the same location. Preferably a room with no natural sunlight and no outside noise being introduced.
4. *Learning Effects* - We must run the experiments in such a way that all participants aren't biased by the order they try the interfaces out. More on this in the next section.

Experimental Design Architecture & Counterbalancing

Subject Design Selection:

- For this study, we will be using a within-subjects design with an alternating A -> B, B-> A counterbalancing scheme.
- This means that all subjects will try both interfaces, and half will start with the revised prototype first, then use the original Madison County Elections website design, while the other half will do the opposite - start with the original design and then try the revised prototype.
- This is to help mitigate learning effects and fatigue as confounding variables in the study.

Participant Sampling Plan:

- *Target User Personas*: for this experiment, we will be looking for participants who are at least 18 years old and have some familiarity with navigating websites on a computer. Ideally, we'd like participants to be eligible voters who might realistically need to find early voting information. We should aim to include people with different levels of technical experience to better represent the variety of people who might visit the Madison County Elections website.
- *Inclusion Criteria*: 18+ years old, comfortable with using a computer
- *Exclusion Criteria*: Participants who have already used the Madison County Elections website, especially ones who have used it to find early voting information in the past. Any participants that I've already used for the Usability Audit or any other assignment in this class already.
- *Sample Size*: We will recruit 10 participants total. All 10 participants will test both interfaces. Since this will be a within-subjects experiment, 5 of them will test in the order of interface A -> interface B, while the other 5 will test in the order of interface B -> interface A

- *Sample Size Rationale*: 10 participants should provide enough observations for our study while remaining realistic in terms of time and recruitment. More participants are always better, so we shouldn't necessarily take the results of the study with 10 participants as 100% fact, but it should give us a good basis to consider.

Ethical Protocol Safeguards

Informed Consent & Ethics:
Thank you for volunteering to participate in this study.

The purpose of this study is to compare two versions of an election information website to understand how effectively users can locate early voting information

You will be asked to complete the same general task using two different interface designs. During each task, I will record how long it takes to complete, any navigation errors you may have, and your rating of how easy or difficult the task felt.

Please understand that this study is evaluating the website interfaces, not your intelligence, technical ability, or performance. There are no penalties for making mistakes or being unable to complete the task.

Your participation is voluntary and you may stop the study at any time, skip questions you don't want to answer, or request that your responses not be included. Your responses will be identified using participant codes rather than your name. Individual responses will not be publicly associated with your identity.

The study should take approximately 10-15 minutes. No personal voter registration information, political leanings, or other sensitive information will be requested of you.

Do you understand the purpose of the study and voluntarily agree to participate?

Yes, I understand the purpose of the study and voluntarily agree to participate.

Signature: _____

Post-Mortem Reflection

1. Why is it critical in HCI empirical research to distinguish between statistical significance ($p < .05$) and practical significance (real-world clinical or operational impact) when evaluating interface performance metrics?

- It is important to distinguish between statistical significance and practical significance because a measurable difference between two interfaces doesn't necessarily mean that the difference is/will be meaningful to the average user. For example, in the above experiment, imagine that participants using my redesigned early voting interface completed the task an average of .5 seconds faster than participants using the original interface. Even if this difference were statistically significant, .5 seconds probably isn't that much of a meaningful improvement if people only visit this website every 2-4 years, if that.
- However, if my redesigned interface reduced the average task completion time by 30 seconds and reduced navigational errors, that would represent a much more substantial usability improvement in the real world.
- Also, if it's something A LOT of people use everyday, maybe multiple times per day, like say Amazon checkout, and the improvement is .5 seconds, that could add up to a lot of time saved over the long haul. Or, say some interface on an EMT's device caused a 2 second delay due to cognitive overload, that could potentially be life threatening to their patients.
- Ultimately, I think deciding which type of significance matters and if either matters more really depends on the interface or user experience you are testing and how many people use it/how frequently they use it/how important it is for it to be efficient in daily life.

2. ***If you chose a Within-Subjects design****, what specific asymmetric transfer or carryover effects could occur between interface conditions, and how does your counterbalancing matrix mathematically isolate that bias?*

- I did in fact choose a within-subjects design, and therefore every participant in my experiment will use both the original Madison County Elections interface (A) and my redesigned prototype interface (B).
- If going interface A then interface B, since the list of early voting locations is the same, a participant might become familiar with the different voting locations being presented and be abnormally better at parsing for their preferred location on the second interface. They could potentially complete the second task faster because they already know what they're looking for.
- It's also possible that if going interface B then interface A, a participant could be more used to the idea of the dropdown menu and might receive more cognitive overload from the long list of locations in interface A than if they hadn't had prior exposure with interface B.

- To reduce this bias as much as possible, I am splitting my participants in half and having one half go in the order of interface A to B, and the other half go in the order of interface B to A.
- Although I doubt this counterbalancing will completely eliminate asymmetric carryover effects, it will at least prevent one interface from always benefitting from being tested second. This should make our comparisons between the two interfaces more fair and help us distinguish actual interface improvements v.s. differences caused by testing order.